

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10 - SUB-STRAND 1.2: THE CELL - FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Driving question: What makes up an organism, and how do the tiny structures inside a single cell keep a whole living thing, from a sukuma wiki plant to a human being in Nairobi, alive? Remember the Kisumu learner who left a sukuma wiki leaf in the sun. It wilted and died. Under the school light microscope, the learner saw box-like compartments with green granules. Under the electron microscope, the same leaf cell showed folded membranes, dark oval bodies and a large fluid-filled space. Use what you observed and learned in Lessons 1 to 8 to answer each question below in your own words. Write your answers in the space under each prompt. Use correct science words and give reasons for your ideas. Where a diagram is asked for, draw and label it neatly. You may use your cell model, your notes and the Driving Question Board.

Part 1: Two microscopes, two pictures

The Kisumu learner was puzzled that the light microscope and the electron microscope showed such different pictures of the same sukuma wiki leaf cell. (a) Explain why the two microscopes show different levels of detail. Refer to resolution and magnification. (b) Name two other differences between a light microscope and an electron microscope. (c) Which structures were visible only under the electron microscope in the sukuma wiki cell?

(a) The two microscopes reveal different levels of detail of the same cell because they have different resolution. Resolution is the ability to show two close points as separate. The light microscope has a resolution of about 200 nm, so it only shows the larger features, such as the cell compartments and the green granules. The electron microscope has a far better resolution, about 0.1 to 0.2 nm, and a much higher magnification, so it shows the tiny internal structures. They are not different cells. They are different levels of detail of the same cell. (b) Any two of: the light microscope uses a beam of light while the electron microscope uses a beam of electrons; the light microscope can view living specimens while the electron microscope needs specially prepared specimens held in a vacuum; the light microscope is cheaper and more accessible to schools in Kenya while the electron microscope is very expensive. (c) The folded membranes, the dark oval bodies (mitochondria) and the large fluid-filled space (the vacuole) were seen only under the electron microscope, along with other internal detail such as the thylakoid membranes in the chloroplasts.

Part 2: Plant and animal cells

(a) Draw and label a plant cell as seen under the electron microscope. Include at least

(a) A correct drawing shows a rectangular or box-like cell with a thick cell wall on the outside, with the cell membrane just inside it. Labels include: cell wall, cell membrane, cytoplasm, nucleus (with nucleolus), large central vacuole, chloroplasts (with grana/thylakoids), mitochondria, ribosomes, rough endoplasmic reticulum, Golgi apparatus. Label lines must touch the correct

eight labels. (b) State three structures found in the sukuma wiki cell but not in a human cheek cell. (c) State one structure found in the human cheek cell but not in the plant cell.	structure and the drawing must use clear, continuous lines. (b) The cell wall, chloroplasts and the large central vacuole are in the plant cell but not in the human cheek cell. (c) Centrioles (or lysosomes) are found in the human cheek cell but not in the plant cell as shown in our lessons.
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Part 3: What each structure does

Choose any five organelles from the sukuma wiki cell or the human cheek cell. For each one, state its function. Then use Kenyan examples, such as a jiko, ugali or githeri, to explain how at least two of these organelles depend on each other.	Nucleus: holds the genetic instructions and controls the cell's activities. Mitochondria: release energy from glucose (from food such as ugali and nyama choma) for the cell. Ribosomes: build proteins. Chloroplasts: contain chlorophyll and use sunlight, water and carbon dioxide in photosynthesis to make glucose. Large central vacuole: stores water and creates turgor pressure against the cell wall. (Also acceptable: Golgi apparatus packages and dispatches materials; lysosomes break down waste; cell membrane controls what enters and leaves; cell wall gives support.) Dependence: chloroplasts make glucose, and mitochondria then release energy from that glucose, like a jiko releasing energy from charcoal. The energy is used by ribosomes to build proteins, which the rough ER and Golgi apparatus send to where they are needed. So the organelles work together like a production line, and if one stops, the others are affected.
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Part 4: Why the leaf wilts and dies

Use what you know about organelles to explain, step by step, why the sukuma wiki leaf left in the sun wilts and eventually dries and dies. Also explain why a fresh leaf is green and feels crisp.	A fresh leaf is green because its cells are packed with chloroplasts containing chlorophyll. It feels crisp because the large central vacuole is full of water and pushes the cell membrane against the rigid cell wall. This pressure is called turgor pressure, and it gives the leaf its shape. When the leaf is left in the sun, it loses water. The vacuole shrinks and turgor pressure falls. The cell wall alone cannot hold the leaf up, so it wilts. Without water the chloroplasts cannot carry out photosynthesis, so less glucose is made. The mitochondria then have less glucose from which to release energy, so the cell cannot power its life processes, and the organelles stop working. The cells die, then the tissues, and finally the whole leaf dries and dies.
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Part 5: From organelle to organism

(a) Arrange these levels of organisation from the smallest to the largest: organ, organelle, organism, cell, organ system, tissue. (b) Give one Kenyan example for each of the sukuma wiki plant and a human being, showing all six levels. (c) Explain why no single organelle or cell can keep a whole organism alive. Then answer	(a) Organelle, cell, tissue, organ, organ system, organism. (b) Sukuma wiki plant: chloroplast, leaf cell, mesophyll tissue, leaf, shoot system, whole plant. Human being: mitochondrion, muscle cell, muscle tissue, stomach, digestive system, human being. (c) Each level can do things the level below cannot do alone. An organelle cannot live without a cell, a cell cannot form a leaf or a stomach by itself, and so on. It is the organised, working together of organelles in cells, cells in tissues, tissues in organs and organs in organ systems that keeps the whole organism alive. Answer to the driving question: An organism is made of cells that are organised into tissues, organs and organ systems. Inside each cell, organelles such as the nucleus, mitochondria, ribosomes, chloroplasts, vacuole and cell wall each do a particular job, and they work together to keep the cell alive. When the cells of a sukuma wiki plant or of a person in Nairobi work together, the whole organism stays alive.
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the driving question in two or three sentences.	
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RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Explaining the two microscopes	Clearly links the different pictures to resolution and magnification, using the correct values or comparisons, and names two further differences correctly.	Says the electron microscope shows more detail and gives one or two correct differences, but without clear reference to resolution.	States that one microscope is better, with no correct scientific reason or differences.
Cell structure and diagram	Accurate, neat labelled plant cell with eight or more correct labels and correct plant versus animal differences.	Mostly accurate diagram with five to seven correct labels and at least two correct differences.	Diagram is incomplete or has several wrong labels and the differences are mostly incorrect.
Organelle functions and interdependence	Gives correct functions for five organelles and explains how at least two depend on each other, using Kenyan examples.	Gives correct functions for most organelles and mentions a link between them.	Gives few correct functions and does not show how organelles work together.
Explaining the phenomenon	Gives a full, correctly ordered cause-and-effect chain from water loss and turgor to photosynthesis, mitochondria and death of cells, tissues and the leaf, and explains the green colour and crispness.	Explains wilting using turgor or photosynthesis, with some steps missing.	Only says the leaf dies because it dries, with no reference to cell structures.
Levels of organisation and answering the driving question	Orders all six levels correctly, gives correct examples for both organisms, and answers the driving question using evidence.	Orders the levels with one or two errors and gives a partly complete answer.	Levels are mostly wrongly ordered and the answer to the driving question is missing or unrelated.

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Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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